

CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

Integrated Systems Laboratory (ETH Zürich)

Zexin Fu, Diyou Shen

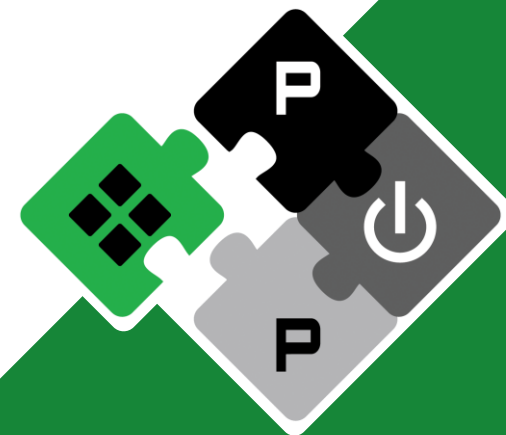
zexifu, dishen@iis.ee.ethz.ch

Alessandro Vanelli-Coralli
Luca Benini

avanelli@iis.ee.ethz.ch
lbenini@iis.ee.ethz.ch

PULP Platform

Open Source Hardware, the way it should be!



@pulp_platform



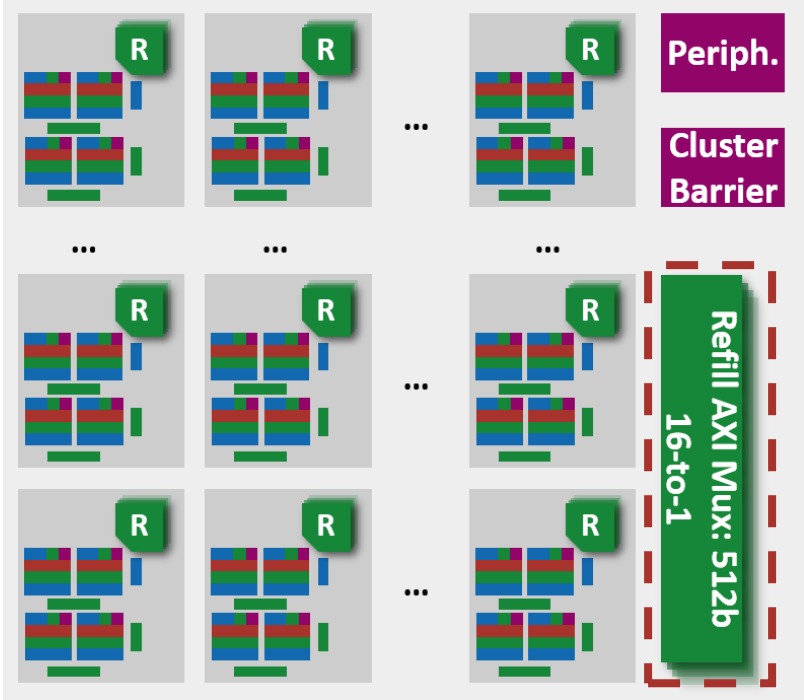
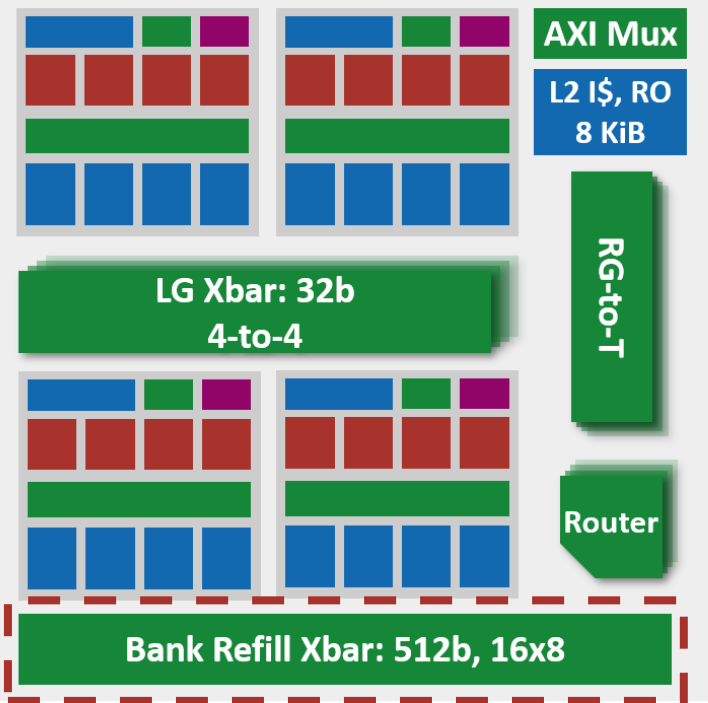
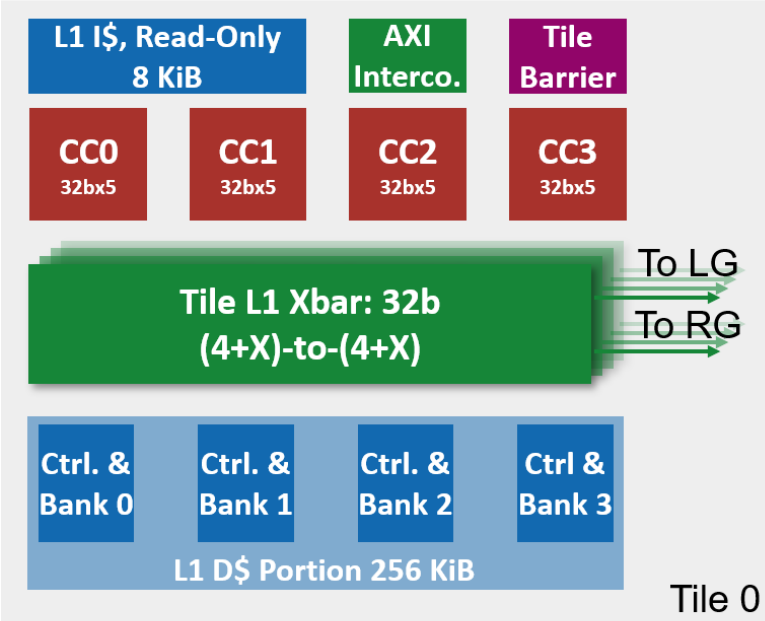
pulp-platform.org



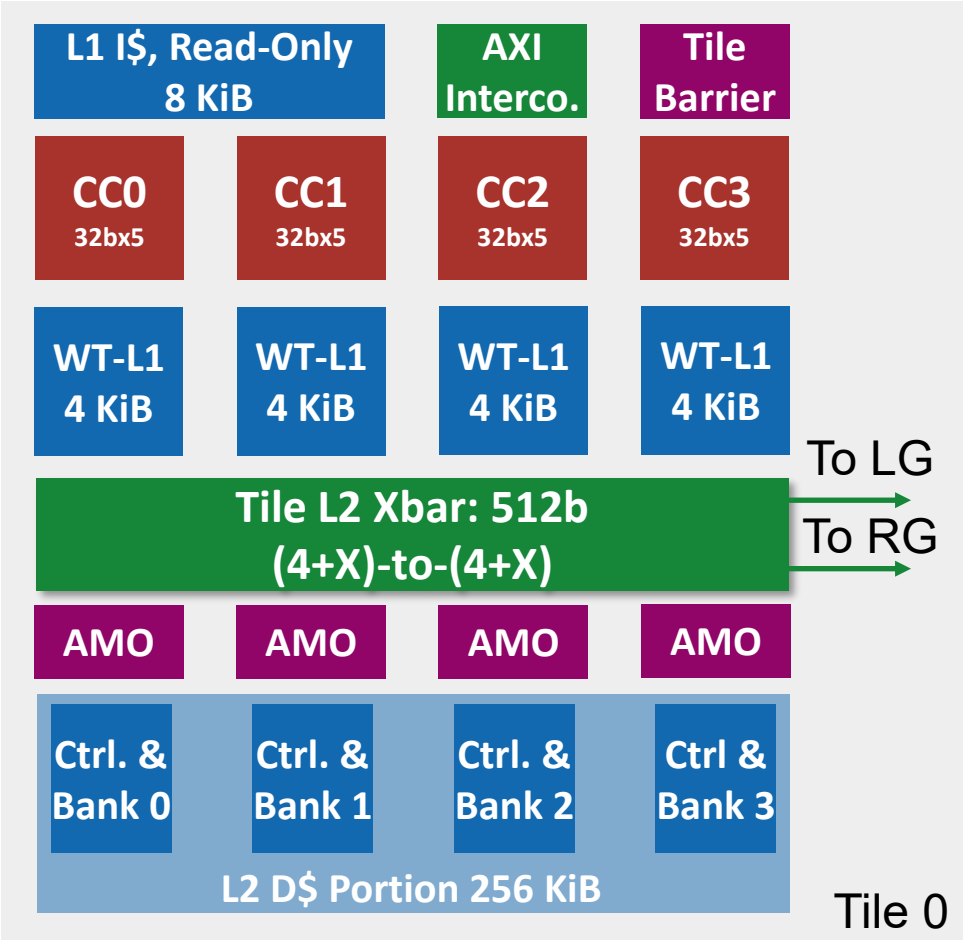
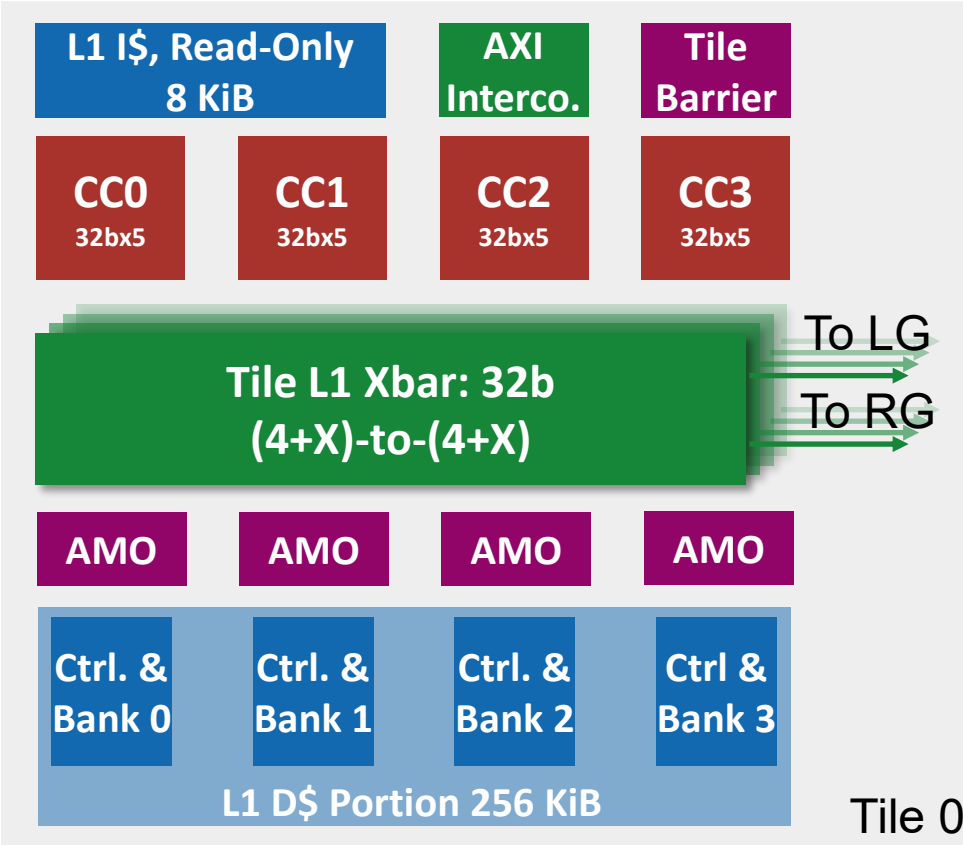
youtube.com/pulp_platform



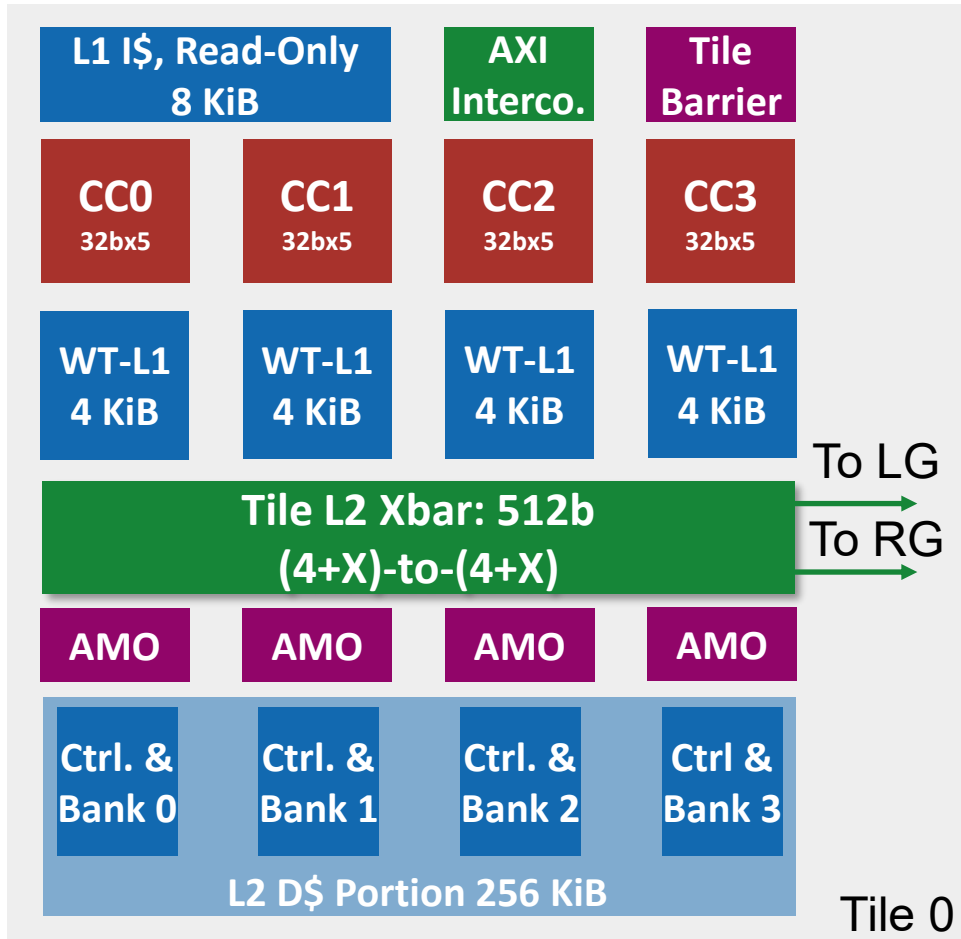
Current Hardware Architecture



Tile-Level Change



Coherence Policy

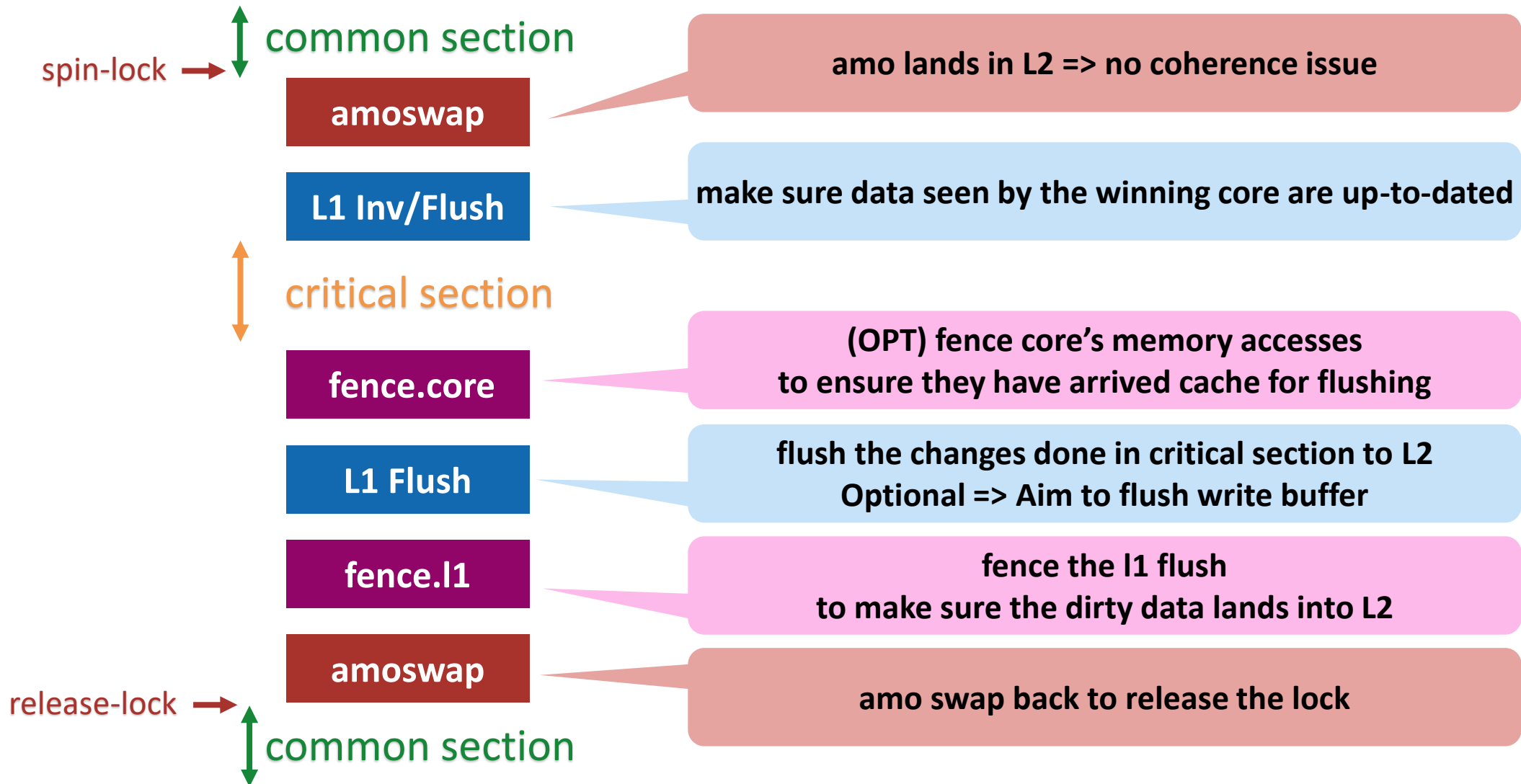


- **Coherence**

- No hardware coherence
- WT-L1, small size $\leq 4\text{KiB}$
- AMO (used for spin-lock) will land to L2
- Plus fence, barrier, and flush



Software Coherence: Lock Routine



Thank you!

Q&A

